

Specific Average Angular Momentum Increase of the Milky Way and Andromeda’s Merged Dark Matter Halo Remnant

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ABSTRACT

Dark matter halos are essential structures in galaxy evolution that govern the dynamics that shape how galaxies form, interact and change over time. Understanding how angular momentum redistributes during halo mergers is crucial, as it influences galaxy morphology, rotation, and overall structure. This project uses an N-body galaxy merger simulation to examine how the specific average angular momentum of dark matter halo particles evolves during a major merger, focusing on the pre-merger Milky Way and Andromeda halos and their combined post-merger remnant, Milkdromeda. The core question explored is whether specific average angular momentum increases in magnitude and shifts radially outward after a merger event. Results show that Milkdromeda’s dark matter halo exhibits significantly higher specific angular momentum at a greater virial radius compared to the pre-merger galaxies, with a peak value nearly four times that of the Milky Way or Andromeda. These findings support theoretical expectations that major mergers drive angular momentum to the periphery and enhance overall halo rotation. This improved understanding of angular momentum transfer during mergers advances our broader knowledge of hierarchical galaxy formation and the dynamical role of dark matter in shaping cosmic structures.

Keywords: Galaxy evolution — Dark Matter Halo — Major Merger — Virial radius — Cold Dark Matter Theory — Local Group — Spiral Galaxy

1. INTRODUCTION

According to **Cold Dark Matter Theory**, most of the universe’s matter consists of slow-moving, non-interacting particles that clump together to form large-scale structures within the universe. The theory predicts that dark matter collapses under gravity to form dense, stable structures known as dark matter halos. A **dark matter halo** is a gravitationally bound, virialized structure composed of dark matter that has decoupled from the universe’s expansion. These halos form in regions with above-average dark matter density and serve as the gravitational wells where baryonic matter collects to form galaxies (G. Besla, private communication, May 6, 2025). Halos play a critical role in shaping the outcome of galactic mergers, such as for merging dark matter halo components of the Milky Way (MW) and Andromeda (M31) galaxies, which will eventually result in a single, uniform halo.

In terms of **galaxy evolution**, the process by which galaxies change and develop over time due to star forma-

tion, interactions with other galaxies, and accumulation of gas and dark matter, determining the post-merger physical attributes of the dark matter halo within the combined MW-M31 system is critical for understanding how galaxies merge and behave dynamically. Examining specifically how the dark matter halo components interact during a merge may also help to shed light on the nature of dark matter and its central role within galaxy formation and internal galactic interactions. Studying how dark matter halos interact through the merging process helps astronomers piece together the evolution and future properties of these two galaxies as they become one, which results in a better understanding of galaxy merger dynamics outside of our galactic **Local Group** (our gravitationally bound group of nearby galaxies). Further understanding of dark matter also has broader implications for particle, nuclear, and other areas of contemporary physics.

Regarding dark matter halo dynamics and its role in galaxy evolution, the literature states that dark matter halos are essential galactic structures that surround/host galaxies (J. Dubinski 1992). Galaxies form at the center of dark matter halos, and dark matter halo

development/evolution is consistent in galaxies across the universe (C. Frenk & S. White 2012). Therefore, dark matter plays a significant role in shaping all galaxies, regardless of their size or brightness (J. Wolf et al. 2010). Initial central galaxy formation affects the shape and mass of the surrounding dark matter halo because as baryons (normal matter) gather at the center of a halo, the shape of the halo becomes more symmetric and less elongated (M. G. Abadi et al. 2010). Dark matter halos form and grow, starting as small structures that stop expanding, collapse and then merge or gather more dark matter to become larger, in a process known as hierarchical growth (C. Frenk & S. White 2012). Merging halos of similar size, such as MW and M31, will thoroughly mix their layers (C. Frenk & S. White 2012). Final merged halos can be elongated or flattened, and the size increases in a predictable way based on the energy and spin from before the merger (N. E. Drakos et al. 2019). Dark matter also has significant implications for a galaxy's angular momentum. Angular momentum is transferred from central regions to the halo's outer edges during galactic mergers (S. Peirani et al. 2004). Therefore, a significant increase in angular momentum at greater distances after a merger is expected. So, specific average angular momentum should be greater at further radii for merged galaxies than compared to the pre-merger galaxies. Dark matter halos are also categorized into two types: "accretion" halos, which grow smoothly, and "merger" halos, which experience immediate growth by quickly absorbing large amounts of mass. Over time, all halos increase their spin, but halos formed through mergers gain more spin relative to their size than those formed through steady accretion (S. Peirani et al. 2004). When two galaxies merge, they also don't just combine their internal spin, they bring in orbital angular momentum from their paths around each other. This extra angular momentum is transferred into the final dark matter halo, meaning it will spin faster than either of the original halos J. S. Bullock et al. (2001). Finally, the shape and rotation of cosmic halos are influenced by the direction and strength of tidal forces, which primarily cause halos to spin smoothly along specific axes, helping to explain galaxy rotations (J. Dubinski 1992). The Milky Way and Andromeda are on a nearly head-on collision, but they still have a slight impact parameter (they aren't perfectly aligned). Even small "misses" can generate torques that increase the spin of the resulting halo H. Hetzner & A. Burkert (2006). Also, a spin parameter (λ), which tells you how much a dark matter halo is spinning compared to how much it could be spinning, is positively correlated with specific angular momentum. So, by calculating the specific

angular momentum we can understand how much faster a merged galaxy is spinning or rotating. Higher λ s represent more rotation M. Vitvitska et al. (2002). Overall, dark matter halos are the static framework for galaxy creation and are the ultimate determining factor for galaxy's physical and dynamical attributes, predominantly angular momentum and spin.

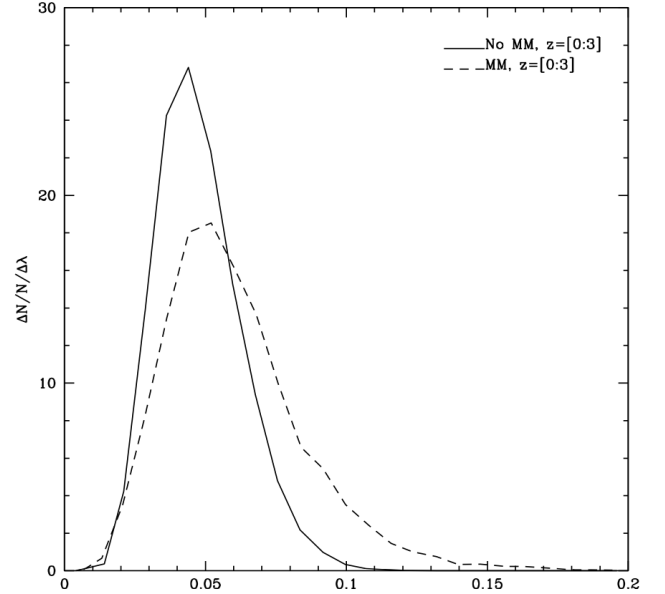


Figure 1. Figure 10 from Ref. (M. Vitvitska et al. 2002) This graph shows how dark matter halos that went through a major merger (dashed line) tend to spin faster than those that didn't (solid line). The dashed line demonstrates higher spin parameter. This rise in spin is a consequence of increased specific angular momentum transferred during mergers, particularly through tangential orbits, as specific angular momentum and the spin parameter are directly correlated. Understanding this angular momentum buildup is essential for explaining why post-merger galaxies often rotate faster.

Several open questions still exist regarding dark matter halo mergers, which are crucial for advancing our understanding of galaxy formation and structure. Key inquiries focus on the specific properties of dark matter that influence merger dynamics and the detailed physical processes involved, including the transfer of angular momentum (J. S. Bullock & M. Boylan-Kolchin 2017). Questions remain about the effects of baryonic matter and feedback mechanisms from star formation and black hole activity on these mergers (J. S. Bullock & M. Boylan-Kolchin 2017). In addition, understanding the time it takes for the merged dark matter halos to reach equilibrium is a significant area of ongoing study. These questions highlight the complexity of halo mergers and their fundamental role in cosmological research.

2. PROJECT

This project investigates the dynamics of the merged dark matter halo versus what the MW halo and M31 halo dynamics were individually before the merger. I will determine the specific average angular momentum of MW, M31 and the merged Milkdrameda dark matter halo at their virial radii graphically to observe how and where average angular momentum peaks for each. I will observe how the specific average angular momentum differs before the merger, when MW and M31 are individual galaxies, and when the galaxies combine to form Milkdrameda through plotting linear fits. These three results will be compared to explore how the profiles differ.

Overall, this project addresses questions about the physical processes involved in a dark matter halo merger, and how the MW-M31 merger affects specific average angular momentum of the halo particles. Specifically, I am investigating if the specific average angular momentum as a function of galactic radius will increase post-merger? And also, is the peak specific average angular momentum at the same or different radii compared to the peak average angular momentum of the dark matter halos in either galaxy before they merged?

Answering questions about the physical processes involved in a dark matter halo merger, specifically regarding how the MW-M31 merger affects the specific angular momentum of the halo particles, is critical for several reasons in the context of galaxy evolution. Studying changes in angular momentum due to galaxy mergers helps us to understand the dynamic processes that drive galaxy evolution. Angular momentum is a fundamental property that affects the shape and structure of galaxies, influencing everything from star formation to the distribution of matter. Investigating how the average angular momentum changes across galactic radii post-merger could help infer how mergers reshape galaxies, such as an increase in angular momentum at further radii might suggest more extended/disrupted galactic disks, impacting the overall structure of the merged galaxy. The location of the peak average angular momentum, depending on if it shifts or remains the same post-merger, provides insights into the effectiveness of theoretical models in predicting the outcomes of **major merger** events, or the collision between similarly sized galaxies. Also, since dark matter constitutes a significant portion of a galaxy's mass and influences its gravitational field, understanding its role during mergers could reveal more about the nature of dark matter.

3. METHODOLOGY

This project utilizes simulation details from literature by Besla +2012 to investigate changes in specific average angular momentum pre and post merger (R. P. van der Marel et al. 2012). The N-body simulation data in this paper models the gravitational interactions of multiple bodies, such as galaxies and different particle types within each galaxy, to understand their dynamics and evolution over time. Before these two galaxies are merged, initially, the Milky Way is a barred **spiral galaxy** with a central bulge, a prominent stellar disk with spiral arms with a diverse range of stars/matter, and a surrounding, elongated dark matter halo. Andromeda is similar in size to the Milky Way, as it is also a barred spiral galaxy with a central bulge, surrounded by a large stellar disk with spiral arms rich in young stars, gas, and dust. It also has an extensive dark matter halo, and ongoing research is still determining its exact halo shape. Both galaxies' dark matter halos are quite extensive and are most likely slightly elongated. Both also have a **virial radius**, or the outer edge of a bound halo, at around 200 kpc where the angular momentum peaks. These two galaxies will merge in around 6.7 Gyrs, and are part of our galactic Local Group, along with other satellite and dwarf galaxies.

The project code will analyze the dynamics of the merged dark matter halo compared to the pre-merger dynamics of the Milky Way (MW) and Andromeda (M31) halos. It selects two points in time, before the merger (Snapshot 0) and post-merger (Snapshot 680), calculates the specific average angular momentum magnitude at the virial radii for MW, M31, and the merged Milkdrameda, and compares these profiles through applying linear fits to understand their differences. From the simulation data, I will only utilize particle type 1, which is dark matter halo particles, all with equal mass. Only the LowRes simulation data is needed for this project.

The functions in my code calculate the center of mass position r_{COM} using the equation for center of mass, where R_i is the position vector of particle i , and N is the total number of particles within a maximum radius, R_{max} . To improve accuracy, the code iteratively reduces R_{max} and recalculates r_{COM} until the value converges, or stops changing. This method ensures a stable center of mass for both pre and post merger galaxies, which is necessary for calculating angular momentum. To compute the absolute specific average angular momentum, L_{ave} , at a given radius r , the code calculates the cross product of position and velocity vectors for particles within a radial bin, relative to the system's center of mass. Each term $(\mathbf{r}_i - \mathbf{r}_{COM}) \times (\mathbf{v}_i - \mathbf{v}_{COM})$

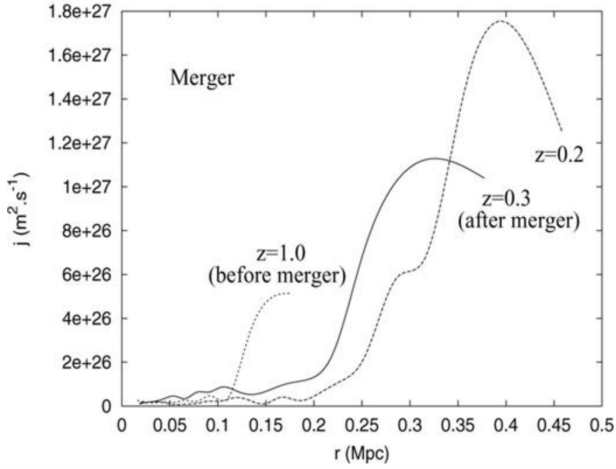


Figure 2. Figure 6 from Ref. (S. Peirani et al. 2004) This graph demonstrates how the average angular momentum of the dark matter halo changes as a function of the radius before and after a merger. The first pre-merger curve represents an arbitrary galaxy’s specific angular momentum as peaking earlier compared to the post merger curve, which peaks at a further virial radius. This indicates that angular momentum is transferred to a merged galaxy’s periphery and generally increases in magnitude once merged, as indicated by the higher peak.

gives the specific angular momentum of particle i , which is averaged over all N particles (type 1) in the bin. This equation in the code computes specific average angular momentum at the galaxies’ virial radii, helping to reveal the rotational structure/dynamical evolution of the dark matter halo.

Three different plots will be generated through the code showing dark matter halo specific average angular momentum as a function of radii for MW and M31 individually pre-merger and for the merged Milky Way halo. These plots are scatter plots, where data points represent the magnitude of specific average angular momentum at a specific radius. From these scatter plots, peak specific angular momentum will be calculated at the virial radii of the galaxies to determine if it is a larger magnitude post-merger. Code from Homework6’s OrbitCom notebook is also imported to determine where the post-merger snapshot is graphically (past 6.7 Gyrs) and the Homework5 Mass Profiles for Milky Way components graph was utilized to determine the radius at which around 80% of the dark matter halo mass is enclosed for the pre-merger galaxies (around 200 kpc). This graph helped to determine the virial radius utilized in the code.

The literature states that the transfer of angular momentum, especially after major mergers, moves from the central regions to the periphery and is larger in magni-

tude (S. Peirani et al. 2004). Therefore, after the dark matter halos of MW and M31 merge into one, I hypothesize that the peak average angular momentum should occur at a further virial radius from the center and will have a greater magnitude (peak) than compared to the two pre-merger, individual galaxies.

4. RESULTS

Figure 3 shows how the specific average angular momentum of dark matter halo particles in the Milky Way evolves as a function of radius before the galaxy merger occurs. The angular momentum is computed in radial bins extending outward from the Milky Way’s center of mass. The resulting scatter plot demonstrates a gradual rise in specific angular momentum with increasing radius, reaching a peak at 3000.98 kpc · km/s at the virial radius (200 kpc) before tapering off after. This reflects the rotational structure of the Milky Way’s dark matter halo prior to the merger. The Milky Way’s specific average angular momentum increases with radius but remains relatively low, particularly beyond 200 kpc.

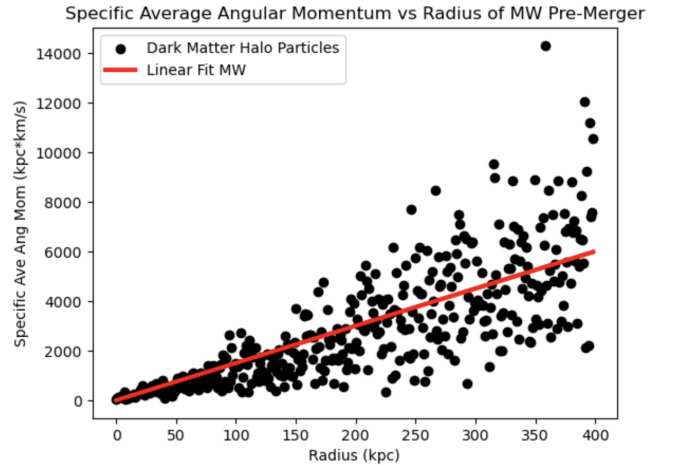


Figure 3. This scatter plot demonstrates the specific average angular momentum of dark matter halo particles in the Milky Way galaxy as a function of radius before the galaxy merger. The plot shows a gradual increase in specific angular momentum with radius, peaking within approximately 200 kpc (at the virial radius) before flattening out, as the scatter points become more horizontal past this point. The specific angular momentum peak is calculated to be 3000.98 kpc · km/s. The overall magnitude remains relatively low beyond the inner regions, reflecting the structure of the pre-merger Milky Way halo.

The Figure 4 scatter plot presents the radial profile of specific average angular momentum for M31’s dark matter halo particles prior to the merger. Like the Milky Way plot, the computation is based on relative positions and velocities with respect to M31’s center of mass at

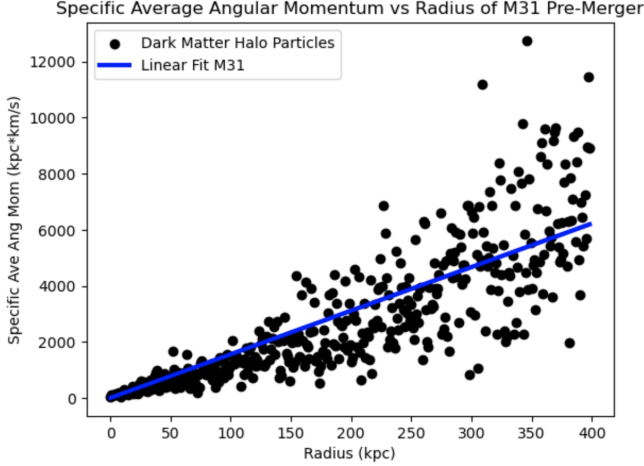


Figure 4. This scatter plot demonstrates the specific average angular momentum of dark matter halo particles in the Andromeda galaxy as a function of radius before the galaxy merger. The plot shows a gradual increase in specific angular momentum with radius, peaking within approximately 200 kpc (at the virial radius) before flattening out, as the scatter points become more horizontal past this point. The specific angular momentum peak is 3111.66 $\text{kpc} \cdot \text{km/s}$. The overall magnitude remains relatively low beyond the inner regions, reflecting the general structure of the pre-merger Andromeda halo.

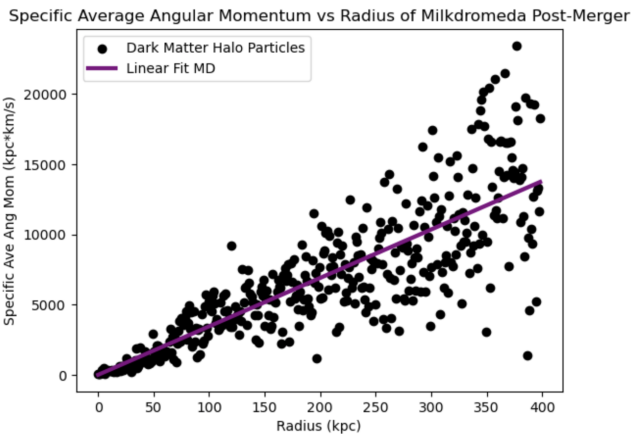


Figure 5. This scatter plot demonstrates the specific average angular momentum of dark matter halo particles in the Milkdromeda galaxy as a function of radius after the galaxy merger. The plot shows a gradual increase in specific angular momentum with radius, peaking within approximately 350 kpc (estimated merged virial radius) before flattening out, as the scatter points become more horizontal past this point. The specific angular momentum peak is around 12071.73 $\text{kpc} \cdot \text{km/s}$. This specific angular momentum is 4 times greater, or 25% larger, than compared to the momentum values for the individual pre-merger galaxies. Therefore, the plot demonstrates that merged halos have a greater specific angular momentum magnitude and at a greater virial radius.

snapshot 0. The angular momentum steadily rises across the inner regions, peaking at 3111.66 $\text{kpc} \cdot \text{km/s}$ at the virial radius 200 kpc. The similarity in profile shape between this and the Milky Way's figure supports the idea that both halos exhibit structured but limited/slower rotation or spin before merging. M31's dark matter halo shows a modest rise in specific angular momentum with radius, but values remain relatively low beyond the central regions.

The post-merger scatter plot, Figure 5, illustrates the post-merger rotational dynamics of the newly formed Milkdromeda dark matter halo, computed at snapshot 680 well after the major merger event. The same analysis was applied by computing average specific angular momentum in radial bins relative to the center of mass of the merged system. Unlike the pre-merger plots, this figure shows a dramatic rise in specific average angular momentum peaking at 12071.73 $\text{kpc} \cdot \text{km/s}$ at a further virial radius of 350 kpc, with values significantly exceeding those seen in either pre-merger halos. The specific angular momentum of Milkdromeda is calculated to be 4.023 times larger, or 24.86% larger than that of the MW. The specific angular momentum of Milkdromeda is also calculated to be around 3.88 times that of the MW, or a 25.78% increase. The difference in slope of the linear fit for Milkdromeda versus the MW is 19.49 km/s , and the difference in slope of the linear fit for Milkdromeda versus M31 is 18.93 km/s . This reflects the increase in magnitude with the steeper slopes and the redistribution of angular momentum outward during the merger process, driven by torques, dynamical friction and additional orbital momentum. Therefore, after the major merger, Milkdromeda's dark matter halo exhibits much higher specific average angular momentum at a greater virial radius, confirming that major mergers significantly enhance outer halo rotation.

5. DISCUSSION

The comparison between Figures 3, 4, and 5 clearly demonstrates how the specific average angular momentum of the MW and M31 dark matter halos evolved as a result of the major merger. Figures 3 and 4 show that, prior to the merger, both galaxies' dark matter halos have only a modest increase in specific average angular momentum with radius, with peaks occurring at similar virial radii around 200 kpc. At this radius, dark matter halo particle density is sufficient enough to call this the angular momentum peak, at around 3000 $\text{kpc} \cdot \text{km/s}$ for MW and M31 individually. In contrast, Figure 5, which represents the post-merger Milkdromeda dark matter halo, reveals a dramatic increase in specific average angular momentum at much larger

radii, with values far exceeding those observed in either the MW or M31 individually. At its further virial radius of 350 kpc, Milkdrameda’s specific angular momentum is around 12000 $\text{kpc} \cdot \text{km/s}$, demonstrating the specific angular momentum is much greater than MW or M31. The specific angular momentum of Milkdrameda is calculated to be around 4 times that of the MW or M31, or a 25 percent increase. Through comparing linear fits, Milkdrameda’s has a steeper slope, also indicating this increased magnitude as a function of radius. This increased magnitude at greater radii strongly supports the hypothesis that major mergers redistribute angular momentum outward, enhancing the rotation of the outer halo regions. These results align with my hypothesis that angular momentum increases in magnitude with radius towards the periphery of the merged galaxy, compared to the individual galaxies pre-merger. Therefore, the merged galaxy has greater overall spin/rotation as predicted.

The results align with theoretical expectations that major mergers transfer angular momentum from the central regions to the outskirts, causing the outer, merged halo to spin faster/exhibit a higher magnitude of specific angular momentum (S. Peirani et al. 2004). This confirms that major mergers produce greater rotation/spin due to increased specific angular momentum M. Vitvitska et al. (2002). Torques and dynamical interactions during the merger explain why the merged galaxy spins more rapidly and achieves greater angular momentum values at larger distances from the center of mass (J. Dubinski 1992). The literature further indicates that the rotation of halos is influenced by the direction and strength of tidal forces (J. Dubinski 1992). Since the Milky Way and Andromeda are not perfectly aligned, small offsets during their merging process generate the torques that contribute to this increased spin H. Hetzner & A. Burkert (2006). Therefore, merged halos spin faster than either of the original halos due to the additional orbital angular momentum introduced during the merger, consistent with what my results demonstrated J. S. Bullock et al. (2001). My findings also support the conclusion that dark matter halos formed through mergers experience an increase in spin relative to their size (S. Peirani et al. 2004). The specific angular momentum quadrupled in size for the merged galaxy compared to the pre-merger systems, reinforcing the idea that the dark matter halos became thoroughly mixed (C. Frenk & S. White 2012). Additionally, the combination of dark matter particles to form the merged halo illustrates the broader concept of how galaxies evolve through hierarchical growth (C. Frenk & S. White 2012). This result shows that major

mergers play a crucial role in redistributing and boosting the angular momentum of dark matter halos, especially at larger radii. It highlights how mergers not only grow galaxies in size but also fundamentally reshape their rotational structure/dynamics, driving the hierarchical evolution of galaxies over time.

However, there are several uncertainties in this analysis. Small errors can arise when calculating the center of mass and relative velocities, and the choice of radial bin size and simulation resolution can influence the measured angular momentum profiles. Excluding baryonic matter and satellite galaxy M33 also means the results do not fully represent the precise galaxy dynamics of this merged halo. I also roughly estimated the virial radii of MW and M31 to both be around 200 kpc and Milkdrameda’s to be 350 kpc using previous class code. This assumption could lead to slightly different angular momentum values as determined by the linear fits. The linear fits were also adjusted to go through the point (0,0). Despite these uncertainties, the results clearly support the idea that major mergers significantly enhance the angular momentum of dark matter halos and play a key role in shaping the structure and evolution of galaxies.

6. CONCLUSION

Dark matter halos are critical to understanding galaxy evolution, as they govern the gravitational dynamics that shape how galaxies form, interact, and evolve over time. This project used an N-body simulation to examine how the specific average angular momentum of dark matter halo particles changes during a major merger, focusing on the Milky Way and Andromeda galaxies and their post-merger remnant, Milkdrameda. The goal was to determine whether angular momentum increases and if the virial radius shifts outward in the merged system.

The analysis found that Milkdrameda’s dark matter halo exhibits a significantly higher peak specific average angular momentum at a greater virial radius (350 kpc), reaching values four times greater than those of either pre-merger halo. This finding aligns with the initial hypothesis and supports theoretical expectations that major mergers redistribute angular momentum to the outer regions of the halo, enhancing overall spin. These results confirm that halo mergers play a key role in reshaping rotational dynamics and advancing our understanding of hierarchical galaxy formation.

Future studies could involve incorporating baryonic physics such as star formation, which were excluded from this analysis, but are known to influence halo structure. Including influence from galaxy M33 may also increase the accuracy of these results. Refining the code to

more precisely determine the virial radius or exploring higher-resolution simulations could also improve results. Further analysis should also test whether these findings hold across different types of galaxy mergers beyond the Local Group.

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2. matplotlib Hunter (2007), DOI: 10.1109/MCSE.2007.55
3. numpy van der Walt et al. (2011), DOI : 10.1109/MCSE.2011.37
4. scipy Jones et al. (2001–), Open source scientific tools for Python. <http://www.scipy.org/>
5. ipython Perez and Granger (2007), DOI : 10.1109/MCSE.2007.53

We respectfully acknowledge the University of Arizona is on the land and territories of Indigenous peoples. Today, Arizona is home to 22 federally recognized tribes, with Tucson being home to the O’odham and the Yaqui. The University strives to build sustainable relationships with sovereign Native Nations and Indigenous communities through education offerings, partnerships, and community service.

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